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Intravital calcium imaging of meningeal macrophages reveals niche-specific dynamics and aberrant responses to brain hyperexcitability

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eLife Assessment

This **valuable** study presents a technically sophisticated intravital two-photon calcium imaging approach to characterize Ca^{2+} dynamics in distinct populations of meningeal macrophages in awake, freely behaving mice. These data are **solid** and suggest that meningeal macrophage calcium activity is tightly linked to anatomical sub-compartments, with potential implications for migraine and neuroinflammatory processes. Despite these strengths and broad relevance to neuroimmunology, several technical and interpretational issues limit the study, which could be addressed to strengthen this manuscript.

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Abstract

The meninges, which envelop and protect the brain, host a dense network of resident macrophages with diverse roles in regulating homeostasis and neuroinflammation. Despite their importance, we have a limited understanding of their behavior in vivo. Many dynamic cellular functions of macrophages involve intracellular Ca^{2+} signaling. However, to our knowledge, virtually nothing is known about the spatiotemporal Ca^{2+} dynamics of meningeal macrophages. We developed a chronic intravital two-photon imaging approach and related computational analysis tools to interrogate meningeal macrophage Ca^{2+} dynamics, at a subcellular resolution, in a novel $\text{Pf4}^{\text{Cre}};\text{TIGRE2.0}^{\text{GCaMP6s/WT}}$ reporter mouse model. Using imaging in awake mice, we characterized the Ca^{2+} activity of meningeal macrophages at steady state and in response to cortical spreading depolarization (CSD), an aberrant pro-inflammatory brain hyperexcitability event, implicated in migraine, traumatic brain injury, and stroke. In homeostatic meninges, macrophages in the dural perivascular niche exhibited several Ca^{2+} dynamic features, including event duration and signal frequency spectrum, distinct from those of in the interstitial, non-perivascular niche. Simultaneous tracking of meningeal macrophage Ca^{2+} dynamics and local vasomotion revealed a subset of dural perivascular macrophages whose activity was coupled to behaviorally-driven diameter fluctuations of their associated vessels. Most perivascular and non-perivascular meningeal macrophages displayed propagating intracellular Ca^{2+} activity as well as synchronized intercellular Ca^{2+} elevations, likely driven by extrinsic factors. In response to CSD, the majority of perivascular and non-perivascular meningeal macrophages showed a persistent decrease in Ca^{2+} activity, while a smaller subset displayed Ca^{2+} elevations. Mechanistically, CGRP/RAMP1 signaling mediated the increase but not the decrease in CSD-mediated Ca^{2+} signaling. Collectively, our results highlight a previously unknown diversity of meningeal macrophage Ca^{2+} dynamics at steady state and in response to an aberrant brain hyperexcitability event linked to neuroinflammation.

Introduction

Resident macrophages are key myeloid immunocytes that play an important role in innate immune surveillance and defense across various peripheral tissues and organs [1–3]. The central nervous system also harbors a large subset of parenchymal macrophages, known as microglia, and several distinct subsets of macrophages localized to the brain's border tissues, including the choroid plexus, perivascular spaces, and the meningeal compartments that cover, protect, and support the brain [4–8]. Macrophages are the predominant immune cell type within the brain meninges, and recent studies have demonstrated the diverse ontogeny, transcriptomic profiles, and immune functions of meningeal macrophages at steady state [5, 8–12] and in several neuropathological conditions [8, 13–16].

Cytoplasmic calcium (Ca^{2+}) signaling underlies a wide variety of cellular homeostatic and inflammatory processes in macrophages [17–23]. In addition to intracellular Ca^{2+} elevation, distinct spatiotemporal dynamics - including oscillation patterns, intracellular propagations, and intercellular synchronization of Ca^{2+} signals - may regulate different macrophage functions during steady state and pathophysiology [20–25]. Despite our increased understanding of the diverse molecular signatures and contributions of meningeal macrophages to homeostasis and neuroinflammation, virtually nothing is known about their Ca^{2+} signaling heterogeneity in both healthy and diseased states.

To comprehensively characterize the Ca^{2+} dynamics of individual macrophages localized to the brain meninges, we combined intravital two-photon Ca^{2+} imaging in a novel reporter mouse line, in which the Ca^{2+} reporter GCaMP6s is expressed in platelet factor 4 (Pf4^+) meningeal macrophages, with an event-based signaling analysis pipeline. Our data reveal several distinct spatiotemporal Ca^{2+} dynamic features in perivascular versus non-perivascular meningeal macrophages, including a unique coupling between the Ca^{2+} signals of dural perivascular macrophages and behaviorally-driven vasomotion of their associated dural vessels at steady state. Furthermore, our data uncover both increases and decreases in Ca^{2+} activity in distinct subsets of meningeal macrophages in response to cortical spreading depolarization (CSD), a pathophysiological brain hyperexcitability event linked to neuroinflammation [26] in migraine, traumatic brain injury, and stroke [27]. Mechanistically, our data suggest the calcitonin gene-related peptide/receptor activity-modifying protein 1 (CGRP/RAMP1) axis uniquely mediates CSD-evoked macrophage Ca^{2+} elevation, indicating a brain-to-meninges neuroimmune signaling pathway involving the activation of meningeal peptidergic sensory neurons.

Results

Characterizing macrophage Ca^{2+} signaling features in homeostatic brain meninges

Previous intravital imaging studies exploring the spatiotemporal dynamics of tissue-resident macrophages have primarily used CX3C motif chemokine receptor 1 (CX3CR1)-based mouse reporter strains [13, 22, 28, 29]. However, other resident monocyte-derived cells are labeled in these reporter mice [30, 31]. Moreover, brain microglia also express CX3CR1 [13, 31–33], limiting the use of these reporter mice for resolving the spatiotemporal subcellular Ca^{2+} dynamics of macrophages localized to the relatively thin meningeal layers covering the brain parenchyma. To systematically characterize meningeal macrophage Ca^{2+} dynamics *in vivo* and avoid contamination from Ca^{2+} signals arising from other resident monocyte-derived meningeal cells and superficial parenchymal microglia, we leveraged recent findings showing the expression of Pf4 in meningeal macrophages but not in parenchymal microglia, [10, 16, 34], and generated transgenic reporter mice expressing the highly sensitive Ca^{2+} indicator GCaMP6s (using TIGRE2.0-based Ai162D mice [35]) in Pf4^+ macrophages (using Pf4^{Cre} mice [36]). Notably, nearly 100% of meningeal macrophages are labelled in Pf4^{Cre} -based reporter mice [34]. Although Pf4 is also expressed by megakaryocytes [36], these cells are absent in the brain meninges [5].

Anesthetic agents impact intracellular Ca^{2+} signaling, including in brain macrophages and other non-excitable glial cells [37–39]. We therefore investigated subcellular Ca^{2+} activity of meningeal macrophages in awake, behaving mice. We imaged meningeal macrophages via a chronic cranial window implanted together with a restraining headpost over the intact dura mater overlying the posterior neocortex. This chronic window approach produces minimal inflammatory responses in the cortex and meninges below the window [40, 41]. After at least 7 days of recovery, mice were gradually habituated across multiple days to head restraint while free to run on a wheel (Fig 1b). We tracked Ca^{2+} transients of GCaMP6s-labeled meningeal macrophages using high-speed two-photon microscopy in 37 fields of view (FOVs) from 7 mice. We first corrected the imaging movies for locomotion-evoked meningeal translational shifts using rigid registration [41]. Movies were then processed using the AQuA2 data analysis platform [42], which implements an unbiased event-based approach to capture spatiotemporal Ca^{2+} event dynamics (Fig 1C). Based on spatial analysis, we assigned events ($n=1361$), with their corresponding Ca^{2+} features, to a given macrophage, with data gathered from a total of 503 cells.

We characterized Ca^{2+} dynamics in two anatomically distinct macrophage populations: perivascular and interstitial non-perivascular ($n=122$; $n=381$, respectively, Fig 2A and Video S1). The vast majority of perivascular macrophages (93.4%, $n=114$) displaying ongoing Ca^{2+} activity were associated with vessels in the dura mater. The two meningeal macrophage subpopulations exhibited several distinct Ca^{2+} activity features. While the total area of Ca^{2+} activity in the peri- and non-perivascular macrophages was similar (Fig 2B), the signal perimeter of perivascular macrophages was significantly higher (Fig 2C) and exhibited a more elongated shape (Fig 2A, 2D). The Ca^{2+} event duration in the perivascular macrophages was also longer compared to the non-perivascular subpopulation (Fig 2F). The peak Ca^{2+} activity level (Max DF/F0, Fig 2E) and event rate (Fig 2G), nonetheless, were similar in the two meningeal macrophage subpopulations.

Distinct Ca^{2+} signal frequency spectra may underlie different biological functions of macrophages [22]. We therefore analyzed meningeal macrophage Ca^{2+} signal waveforms by performing a multi-step signal processing and clustering analysis. We clustered cells based on two distinct patterns of Ca^{2+} activity. Cells in Cluster 1 ($n=40$) exhibited a more noisy-like activity pattern characterized by multiple frequencies, while cells in Cluster 2 ($n=463$) displayed a single dominant frequency at 0.01 Hz (Fig 3A). To further explore these cellular signaling differences, we combined the clustering data with AQuA2-derived features and observed that macrophages in Cluster 1 showed a larger event perimeter and lower circularity and peak magnitude compared to those in Cluster 2 (Fig 3D–F). Cluster 1 cells also had a lower signal-to-noise ratio compared to Cluster 2 cells (Fig 3I), but had differences in terms of event area, duration, or rate (Figures 3C, G, H). There was a significant association between cell cluster (1 vs. 2) and cell type, with Cluster 1 predominantly comprised of perivascular macrophages, and Cluster 2 including primarily non-perivascular macrophages (Fig 3J), further indicating that these two meningeal macrophage subpopulations have distinct Ca^{2+} signaling properties.

Intracellular Ca^{2+} signal propagation underlies diverse cellular functions and has been recently identified in macrophages *in vitro* [21] and in skin-resident macrophages *in vivo* [43]. By assessing the Ca^{2+} signal propagation maps of each event within a determined cell, we identified two distinct patterns of activity: propagating events in which Ca^{2+} signals traveled throughout the entire cell, and stationary events (Fig 4A). Propagating events had varied signal source regions and directionality. Most macrophages (perivascular, 94.3%, $n=115$; non-perivascular, 86.9%, $n=331$) exhibited only propagating events, while a small minority of cells displayed a mix of propagating and stationary events or exclusively stationary activity events (Fig 4B).

Macrophage intercellular communication, including synchronized activity, potentially across connected macrophage networks, has been implicated in maintaining tissue homeostasis and immune function [18, 21, 44, 45]. To investigate meningeal macrophage intercellular interactions, we characterized spatiotemporal relationships between Ca^{2+} events in distinct cells within each FOV. We compared temporal factors, including the relative onset latency ($t_{0A} - t_{0B}$) between different macrophages and the duration of the first occurring event ($t_{1A} - t_{0A}$) (Fig 4C). We also

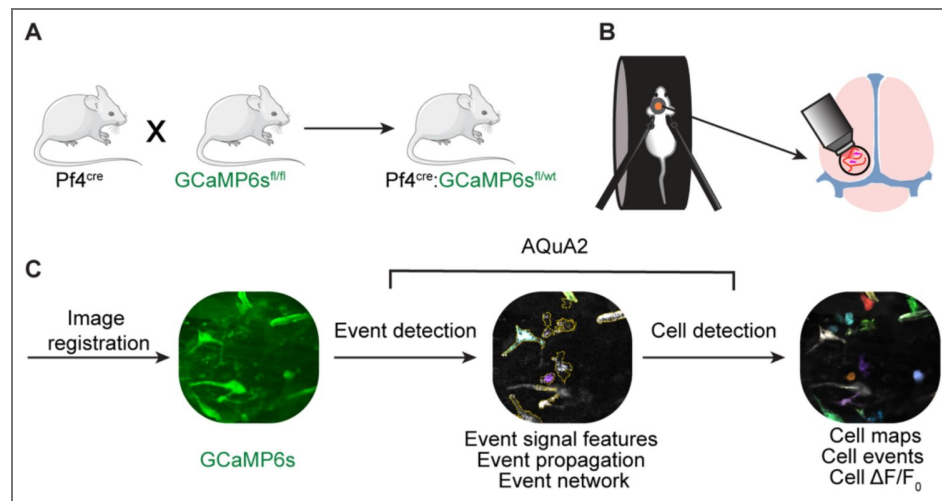


Figure 1. Imaging meningeal macrophage Ca^{2+} dynamics in awake behaving mice.

(A) Pf4^{Cre}:GCaMP6s^{fl/wt} reporter mouse construct for imaging meningeal macrophages Ca^{2+} activity. **(B)** Experimental procedure for two-photon imaging of meningeal macrophage Ca^{2+} activity. Following the implantation of a headpost and a cranial window, mice were habituated to head restraint and subjected to two-photon microscopy while head-fixed on a running wheel to study meningeal macrophage Ca^{2+} activity. **(C)** Macrophage Ca^{2+} imaging processing pipeline.

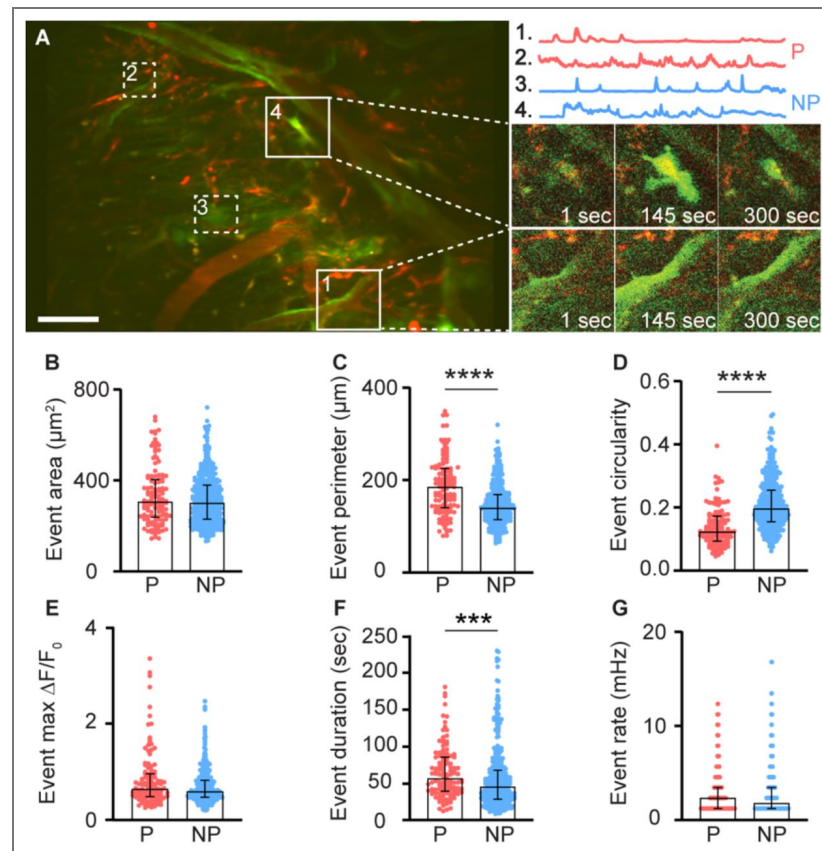


Figure 2. Ca^{2+} dynamic features of meningeal macrophages at steady state.

(A) Left: Mean projection of an example FOV depicting perivascular (P, red; 1, 2) and non-perivascular (NP, blue; 3, 4) meningeal macrophages (white squares). Scale bar 50 μm . Right: Corresponding macrophages with representative 900-second Ca^{2+} activity traces (top) and their fluorescence signal at selected time points (bottom). (B-G) AQuA2-based morphological and Ca^{2+} event functional features of perivascular (P=122 cells) and non-perivascular (NP, n=381 cells) meningeal macrophage. (B) Event area, (C) Event perimeter, (D) Event circularity, (E) Event max $\Delta F/F_0$, (F) Event duration, (G) Event rate. Data (B-G) represent median $\pm$ IQR. *** $p < 0.001$, **** $p < 0.0001$, Mann-Whitney U-test. n=37 fields of view (FOVs) from 7 mice.

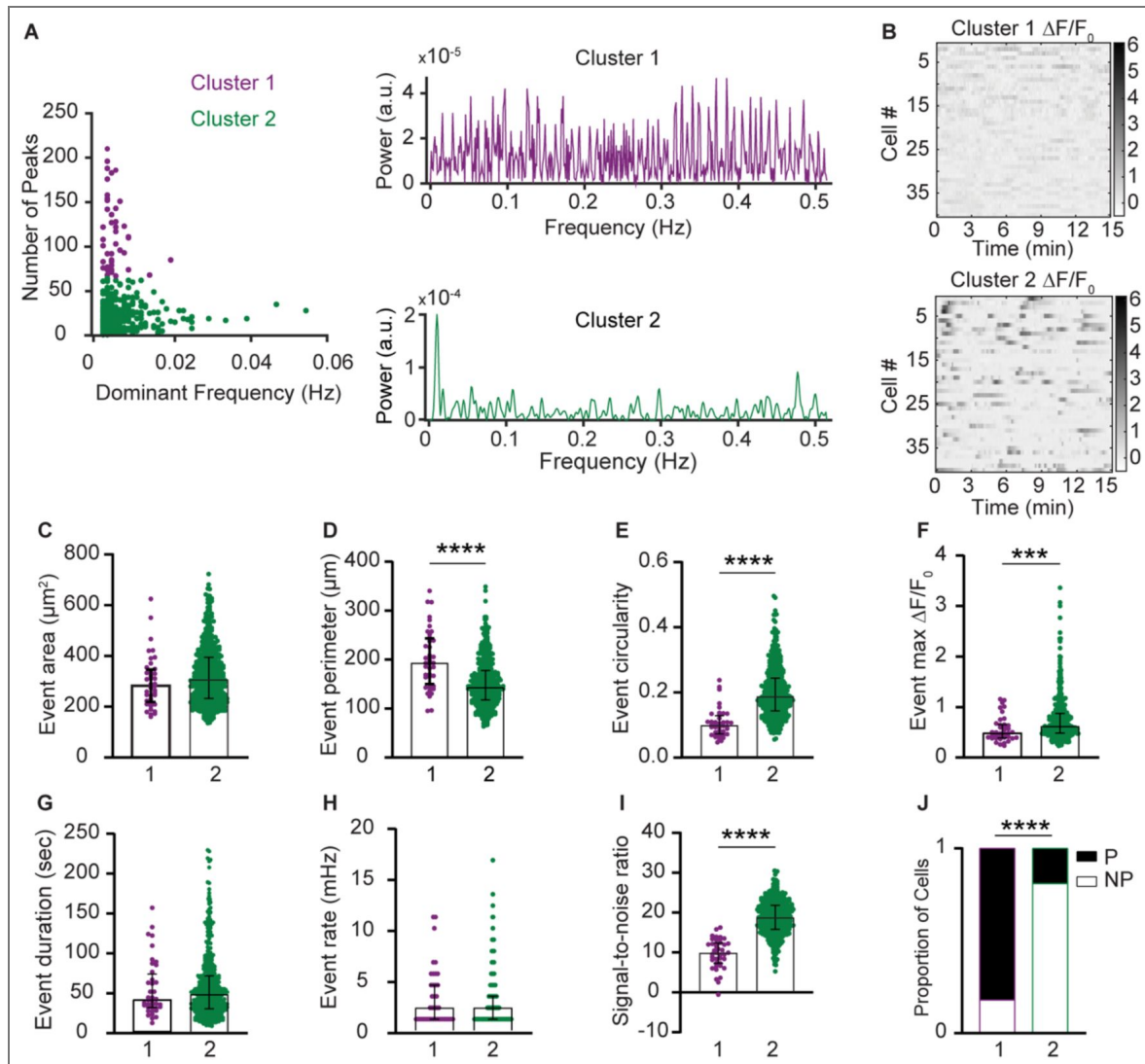


Figure 3. Intracellular Ca^{2+} signal frequency spectra of macrophage subsets in the steady-state meninges

(A) Left: Clustering of meningeal macrophage Ca^{2+} activity based on frequency-domain features and peak detection. Cluster 1 (purple, $n=40$) and Cluster 2 (green, $n=463$). Right: Power spectrum density (PSD) of Ca^{2+} signals for each cluster. (B) Example $\Delta F/F_0$ heatmaps of Cluster 1 and Cluster 2 macrophages. (C-H) Morphological and Ca^{2+} functional features of Cluster 1 and Cluster 2 macrophages. (C) Event area. (D) Event perimeter (E) Event circularity. (F) Event max $\Delta F/F_0$. (G) Event duration. (H) Event rate. (I) Signal-to-noise ratio from Clusters 1 and 2. (J) Distribution of cell types across clusters. Data (C-I) represents median $\pm$ IQR. *** $p < 0.001$, **** $p < 0.0001$, Mann-Whitney U-test Data (J) represents the cell proportion. **** $p < 0.0001$, Chi-square test.

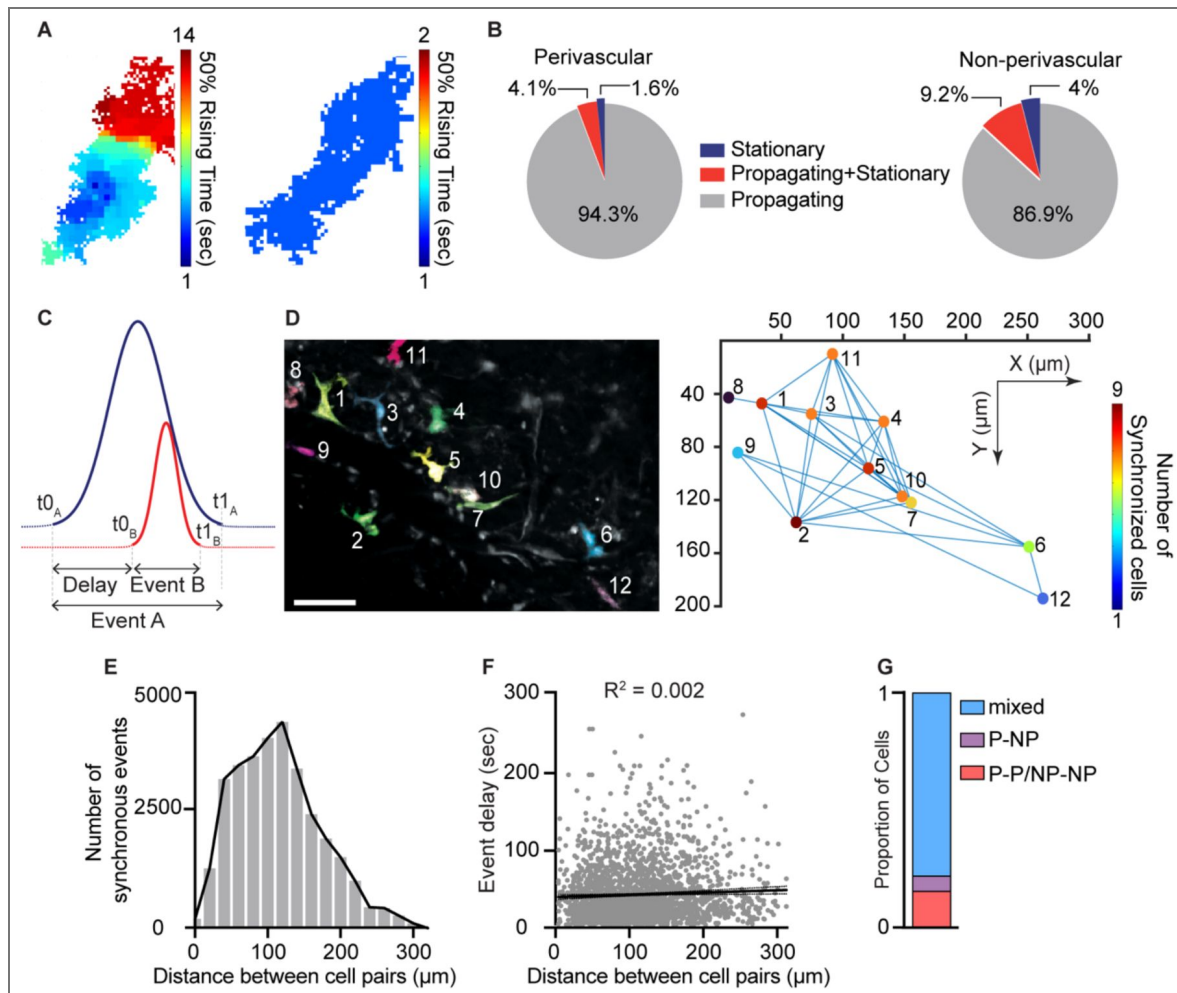


Figure 4. Intracellular propagation and intercellular synchronization of meningeal macrophage Ca^{2+} activity.

(A) Spatial maps of two distinct Ca^{2+} events. Left: Propagating Ca^{2+} activity. Right: Stationary Ca^{2+} activity. **(B)** Distribution of event propagation profiles in perivascular and non-perivascular meningeal macrophages. **(C)** Schematic analysis paradigm for detecting synchronous Ca^{2+} activity in meningeal macrophages. **(D)** Synchronous Ca^{2+} events among meningeal macrophages within a FOV. Left: Mean projection of an example FOV showing Ca^{2+} activity in distinct macrophages (colored/numbered). Scale bar 50 μm . Right: Spatial map of macrophages exhibiting synchronous Ca^{2+} activity. Lines connect macrophages with synchronized Ca^{2+} activity, and colors indicate the extent of Ca^{2+} event synchronization. **(E)** Distribution of distances across macrophage pairs showing different numbers of synchronous Ca^{2+} events. **(F)** Linear regression showing poor correlation between distances of macrophage with synchronized Ca^{2+} activity and event delay. **(G)** Proportion of macrophages exhibiting a specific synchronized interaction (cells interacting only with the same subtype: P-P/NP-NP; cells interacting only with a different subtype: P-NP; mixed (cells interacting with the same and different subtypes)).

compared the distances between macrophage pairs exhibiting concurrent events and the number of synchronous events (Fig 4D [↗](#)). Finally, we calculated the proportion of perivascular and non-perivascular macrophages exhibiting synchronous Ca^{2+} events. Across all FOVs, nearly half of the macrophages (49.3%) exhibited synchronous Ca^{2+} activity, involving cells at a wide range of distances (Fig 4E [↗](#)), suggesting that spatial proximity does not influence event synchronicity. Interestingly, there was poor correlation between the distance of cell pairs exhibiting synchronous events and their timing (Fig 4F [↗](#)), suggesting that an external factor, rather than intercellular coordinated networked activity, mediates this intercellular signaling. When comparing synchronous activity among peri- and non-perivascular macrophages, most cells exhibited mixed interactions between the same and different subtypes (Fig 4G [↗](#)).

Dural perivascular macrophage Ca^{2+} activity is tuned to behaviorally-driven dural vasomotion

Brain border macrophages in the leptomeninges and related parenchymal perivascular spaces regulate pial arterial vasomotion indirectly by affecting vessel stiffness [6]. Yet, interaction between vascular-associated macrophages residing in the dura mater, the outermost meningeal layer, and dural vasomotion remains unknown. We employed a Generalized Linear Model (GLM) approach to investigate functional interaction between dural perivascular macrophage Ca^{2+} signals and behaviorally-driven dural vessel dynamics (Fig 5A [↗](#)). Dural arteries constrict during locomotion, while pial arteries dilate [46]. We analyzed the locomotion-associated responses of 86 meningeal vessels (32 FOVs from 5 mice) and identified a subset (22%, $n=19$), whereby the diameter changes were well fit by a GLM using the locomotion state as a predictor. Of these, we identified 74% ($n=14$) as dural vessels based on their GLM's negative coefficients consistent with constriction (Fig 5B [↗](#) and 5C [↗](#)). Next, we fitted the Ca^{2+} signal observed in perivascular macrophages associated with these dural vessels ($n=35$) to a GLM using the diameter changes as a predictor variable. Overall, the Ca^{2+} activity of 83% ($n=29$) of these dural macrophages was well predicted by the model (average deviance explained across all well-fit macrophages: 0.43 ± 0.15 , mean $\pm$ SD). Analysis of the macrophage-vascular models' beta coefficients revealed two distinct interactions. About half of the macrophages (55%, $n=16$) exhibited negative coefficients (i.e., increase and decrease in Ca^{2+} activity associated with dural vasoconstriction and recovery, respectively; Fig 5B [↗](#) and 5D [↗](#)). The remaining macrophages (45%, $n=13$) exhibited positive coefficients (i.e., decrease and increase in Ca^{2+} activity in response to dural constriction and recovery, respectively; Fig 5B [↗](#) and 5E [↗](#)). The coefficients for increased and decreased macrophage Ca^{2+} activity peaked near zero delay relative to the vasoconstriction and were not statistically different (Fig 5D-E [↗](#) and 5H). These data provide evidence that dural perivascular macrophages are functionally coupled to dural vasomotion, either responding to or mediating such behaviorally driven vascular response.

An acute aberrant pro-inflammatory brain hyperexcitability event drives diverse Ca^{2+} dynamics in meningeal macrophages

CSD is an aberrant brain hyperexcitability event linked to parenchymal inflammation and pain in migraine, traumatic brain injury, and stroke [27, 47, 48], and could also affect meningeal macrophages [49]. In anesthetized mice subjected to an acute CSD episode, a small subset of meningeal macrophages undergoes morphological changes resembling an inflammatory state [26]. Given the direct anatomical and functional connection between the brain and meninges [11, 50] and the involvement of intracellular Ca^{2+} influx in macrophage inflammatory activation [51], we asked whether brain-to-meninges signaling drives intracellular Ca^{2+} elevations in meningeal macrophages in the wake of CSD. We therefore triggered a single CSD episode in the frontal cortex [52, 53] and characterized the related changes in meningeal macrophage Ca^{2+} dynamics. In each experiment, we verified CSD induction based on the associated acute meningeal deformation and/or pial vasoconstriction observed in awake animals ([53] and Video S2). We studied CSD-related Ca^{2+} dynamics in 249 macrophages (perivascular, $n=64$; non-perivascular, $n=185$; 13 FOVs

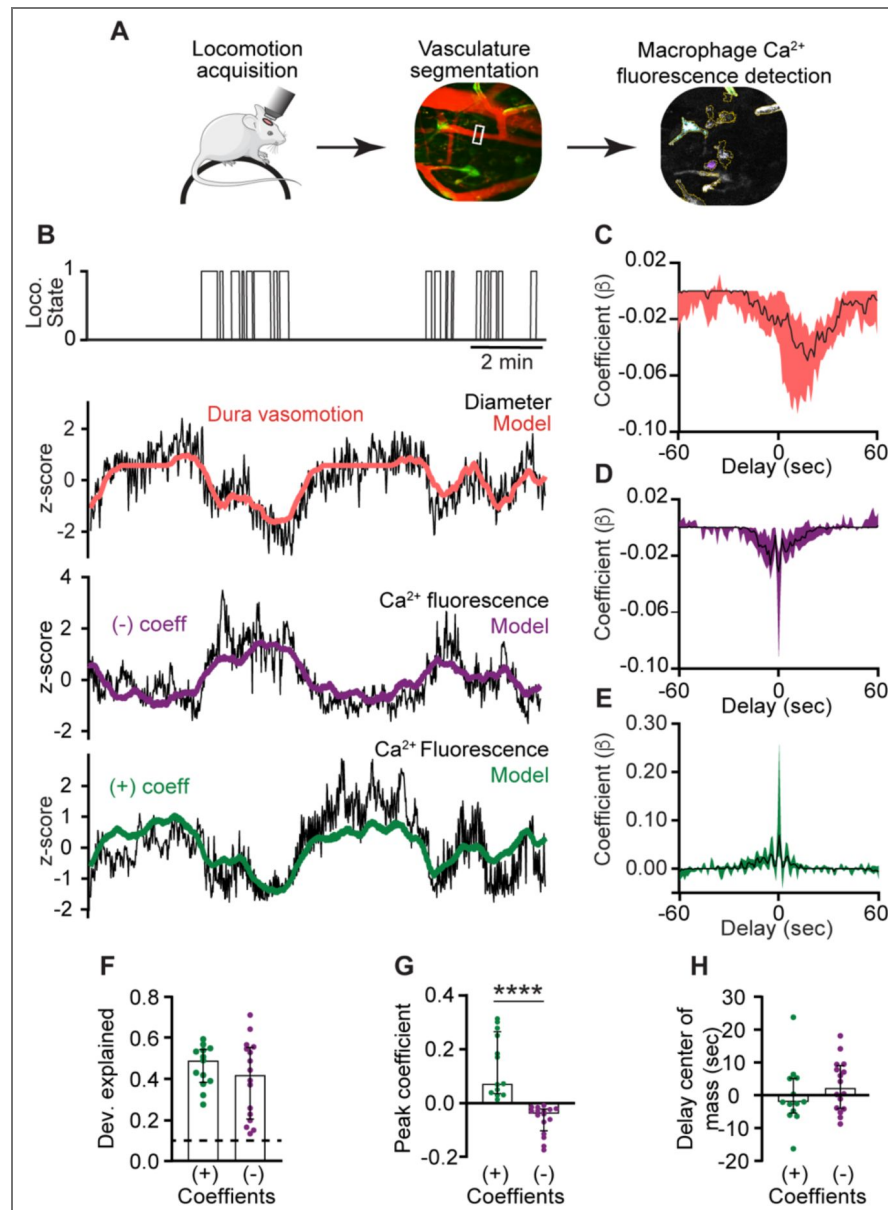


Figure 5. Ca^{2+} signals of dural perivascular macrophages are functionally coupled to behaviorally driven dural vasomotion.

(A) Experimental paradigm: Locomotion data were acquired during imaging in awake-behaving mice. Behaviorally-evoked changes in meningeal vessel diameter were obtained using segmentation of vessels labeled with a tracer during macrophage Ca^{2+} and further tested for coupling with Ca^{2+} signals. (B) Example data of macrophages with Ca^{2+} activity tuned to locomotion-related dural vessel vasomotion. Locomotion bouts (top trace) and dural vessel diameter (black trace) that was well fit by a GLM (red line) using locomotion state as a predictor. Note vasoconstriction during locomotion, indicating a dural vessel. The two bottom traces depict the Ca^{2+} signal (black traces) of distinct meningeal macrophages that were well-fit by a GLM using dural vessel diameter as a predictor, showing either a negative coefficient (purple) or a positive coefficient (green). (C) Temporal profile of dura vessels GLM coefficients averaged across all well-fitted vessels ($n=19$). Traces represent the median across all well-fit ROIs, with shaded regions indicating IQR. (D, E) Temporal profiles of GLM coefficient values for macrophages' Ca^{2+} activity averaged across all well-fitted cells (purple, negative coefficient, $n=16$; green, positive coefficient, $n=13$). Traces represent the median across all well-fit macrophages, with shaded regions indicating IQR. (F) The deviance explained (goodness-of-fit estimates) of the vessel diameter data included in the GLM used to predict macrophage Ca^{2+} activity were not statistically different for macrophages showing negative ($n=16$) and positive ($n=13$) coefficients, suggesting similar interaction levels. (G) Comparison of peak positive and negative coefficients of well-fit macrophage Ca^{2+} activity/vasomotion GLMs. (H) Center of mass of GLM coefficients indicating that dural vessel diameter changes drive bidirectional changes in fluorescence signal at zero delay. Data (F-H) are median $\pm$ IQR. **** $p < 0.0001$ (Mann-Whitney U-test).

from 10 mice). For each cell, we compared Ca^{2+} event rates during the passage of the CSD wave (1 min) and the PostCSD period (30 min) to baseline (PreCSD, 30 min) to assess acute and persistent Ca^{2+} responses, respectively. We characterized cells as exhibiting persistently increased (event rate $> 2 \times$ PreCSD), decreased (event rate $< 0.5 \times$ PreCSD), or unchanged responses. Given the low Ca^{2+} activity rate observed under steady state and the likelihood that no spontaneous Ca^{2+} elevations occur during the brief period of the CSD event, we considered cells to be either acutely activated or exhibiting an unchanged Ca^{2+} response. CSD evoked both acute and persistent Ca^{2+} activity changes in (Fig 6 A-E and Video S2). While smaller subsets of meningeal macrophages exhibited acute (21.3%, $n=53$) and/or persistent increase (22.1%, $n=55$) in their Ca^{2+} activity, the majority of cells (58.6%, $n=146$) exhibited a persistent decrease. An acute increase was observed more often in peri- vs non-perivascular macrophages (perivascular, 32.8%, $n=21$; non-perivascular, 17.3%, $n=32$, Fig 6F). Persistent changes in Ca^{2+} activity were observed similarly in peri- and non-perivascular macrophages (increases; perivascular, 28.2%, $n=18$; non-perivascular, 18.4%, $n=37$; decreases; perivascular, 50.0%, $n=32$; non-perivascular, 61.6%, $n=114$, Fig 6G). The macrophages' propensity to develop a persistent Ca^{2+} increase was unrelated to their acute response (Fig 6H), suggesting that the mechanisms underlying these two temporal responses are distinct. However, cells showing no acute activation were more likely to display decreased Ca^{2+} activity post-CSD (Fig 6H). Finally, we observed that macrophages exhibiting a persistent increase in Ca^{2+} activity had a lower baseline activity when compared to cells showing a persistent (Fig 6I), suggesting that this post-CSD response is influenced by the macrophages' basal response activity.

CGRP/RAMP1 signaling mediates CSD-evoked persistent increase in meningeal macrophage Ca^{2+} activity

Many meningeal macrophages are localized near peptidergic sensory axons [16]. In the wake of CSD, cortex-to-meninges signaling enhances the responsiveness of meningeal sensory neurons and can drive the release of the neuropeptide CGRP from their peripheral nerve endings [52–54]. CGRP-expressing sensory neurons regulate tissue immunity, including meningeal macrophage function, via the CGRP/RAMP1 neuroimmune axis [16, 55]. We therefore asked whether the CSD-related changes in meningeal macrophage Ca^{2+} dynamics we observed involve CGRP/RAMP1 signaling. We pretreated mice with the RAMP1 antagonist BIBN4096 and then imaged meningeal macrophage Ca^{2+} activity (42 cells; perivascular, $n=14$; non-perivascular, $n=28$; 3 FOVs from 3 mice) before and after CSD. As expected, RAMP1 blockade did not affect CSD triggering [56]. When compared to control saline treatment, RAMP1 blockade also did not reduce basal macrophage Ca^{2+} activity (Fig 7B). Blocking CGRP/RAMP1 signaling neither affected the incidence of acute increases in Ca^{2+} activity (Fig 7A and 7C) nor the magnitude of that response (Fig 7D). RAMP1 antagonism, however, inhibited the CSD-evoked persistent increase in the macrophages' Ca^{2+} activity, without affecting the incidence of the persistent decrease (Fig 7E). The data suggest that in the wake of CSD, the CGRP/RAMP1 axis, likely involving a peripheral sensory neuroimmune signaling, is responsible for the prolonged enhancement of Ca^{2+} signaling in a subset of meningeal macrophages, which could potentially mediate their pro-inflammatory response.

Discussion

Resident macrophages in the brain meninges are essential for maintaining brain homeostasis, regulating central nervous system immune surveillance, and mediating neuroimmune responses under pathological conditions [12, 14, 16, 57–59]. Macrophages rely on Ca^{2+} signaling to mediate many of their functions [17–23]. Here, using two-photon microscopy in awake, behaving $\text{Pf4}^{\text{Cre}}:\text{GCaMP6s}$ reporter mice, we describe the heterogeneity of meningeal macrophage Ca^{2+} dynamics at steady state and in response to CSD, an aberrant cortical hyperexcitability event associated with migraine, traumatic brain injury, and stroke. Our data suggest the presence of two subsets of macrophages in discrete meningeal niches, perivascular and non-perivascular, that exhibit several distinct Ca^{2+} signal properties at steady state. We further demonstrate Ca^{2+} activity

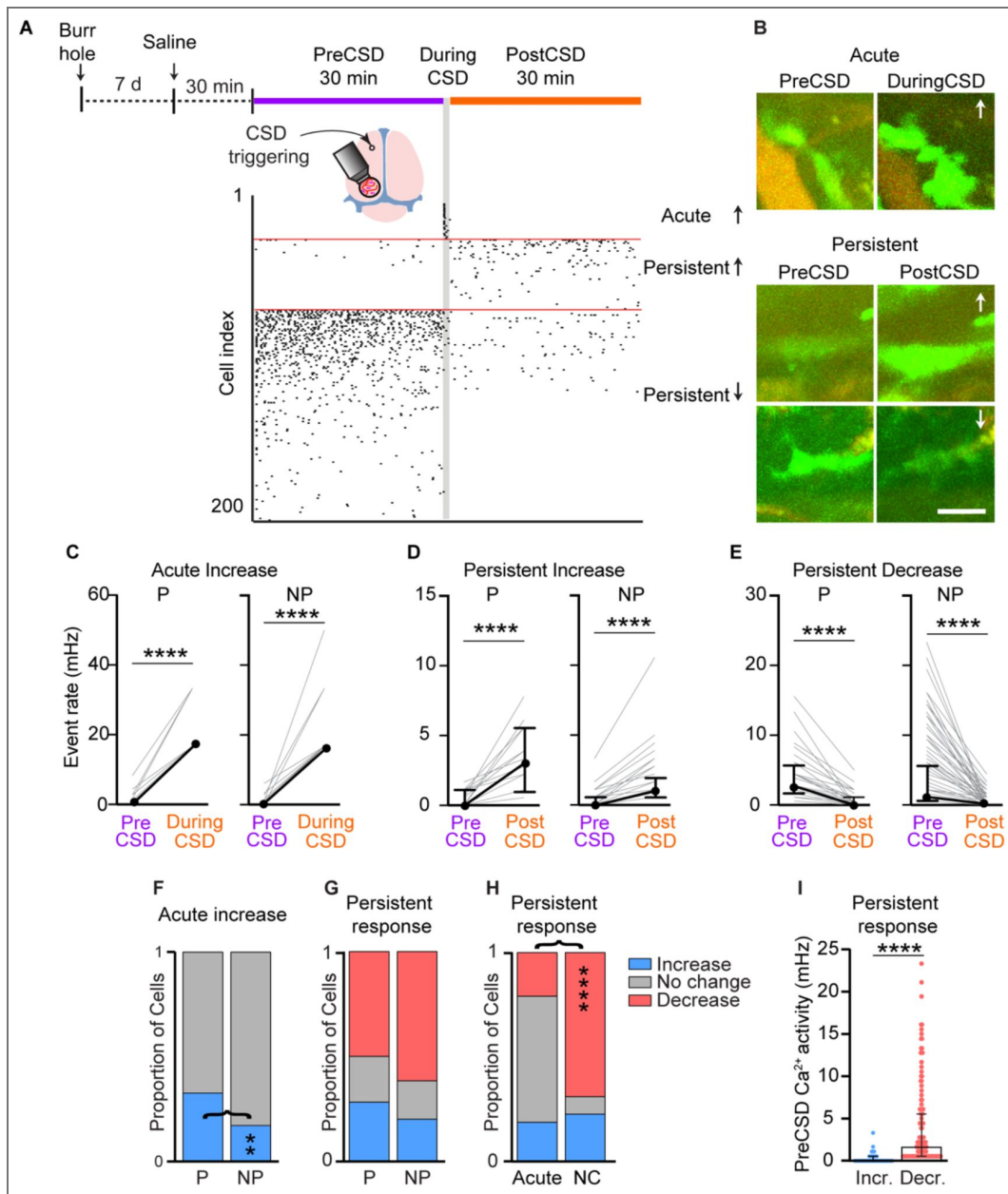


Figure 6. Diverse meningeal macrophage Ca^{2+} dynamics following cortical spreading depression (CSD).

(A) Experimental setup and example data: (top) A small burr hole was drilled above the frontal cortex 7 days before Ca^{2+} imaging. Mice were pretreated with saline 30 min before imaging baseline macrophage Ca^{2+} activity (30 min, PreCSD). CSD was then induced with a pin prick, and macrophage Ca^{2+} activity was assessed during CSD (1 min during CSD), and post-CSD (30 min, post-CSD). (Bottom) A raster plot of macrophage Ca^{2+} activity showing the acute and persistent increases and persistent decrease in response to CSD. (B) Example of macrophage Ca^{2+} fluorescence changes following CSD. Images depict the mean projection over the specific experimental timeline. Arrows indicate an increase or a decrease in Ca^{2+} activity. Scale bar 50 μm . (C) Individual responses of perivascular (P, $n=21$) and non-perivascular (NP, $n=32$) macrophages showing an acute increase in Ca^{2+} activity. (D) Individual response of P ($n=18$) and NP ($n=37$) macrophages exhibiting a persistent increase in Ca^{2+} activity. (E) Individual responses of P ($n=32$) and NP ($n=114$) macrophages showing a persistent decrease in Ca^{2+} activity. (F) Proportion of P and NP macrophages showing an acute increase in Ca^{2+} activity or no acute change. (G) Proportion of P and NP macrophages showing a persistent increase, decrease, or no change in Ca^{2+} activity. (H) Proportion of macrophages displaying distinct persistent responses stratified based on their acute response. (I) Baseline (PreCSD) Ca^{2+} activity in macrophages exhibiting persistent increased ($n=55$) or decreased activity ($n=146$). Data (C-E) are median $\pm$ IQR. **** $p<0.0001$ (Wilcoxon signed rank test). Data (F-H) represents the proportion of cells, ** $p<0.01$; **** $p<0.0001$ (Chi-square test). Data (I) are median $\pm$ IQR. **** $p<0.0001$ (Mann-Whitney U-test).

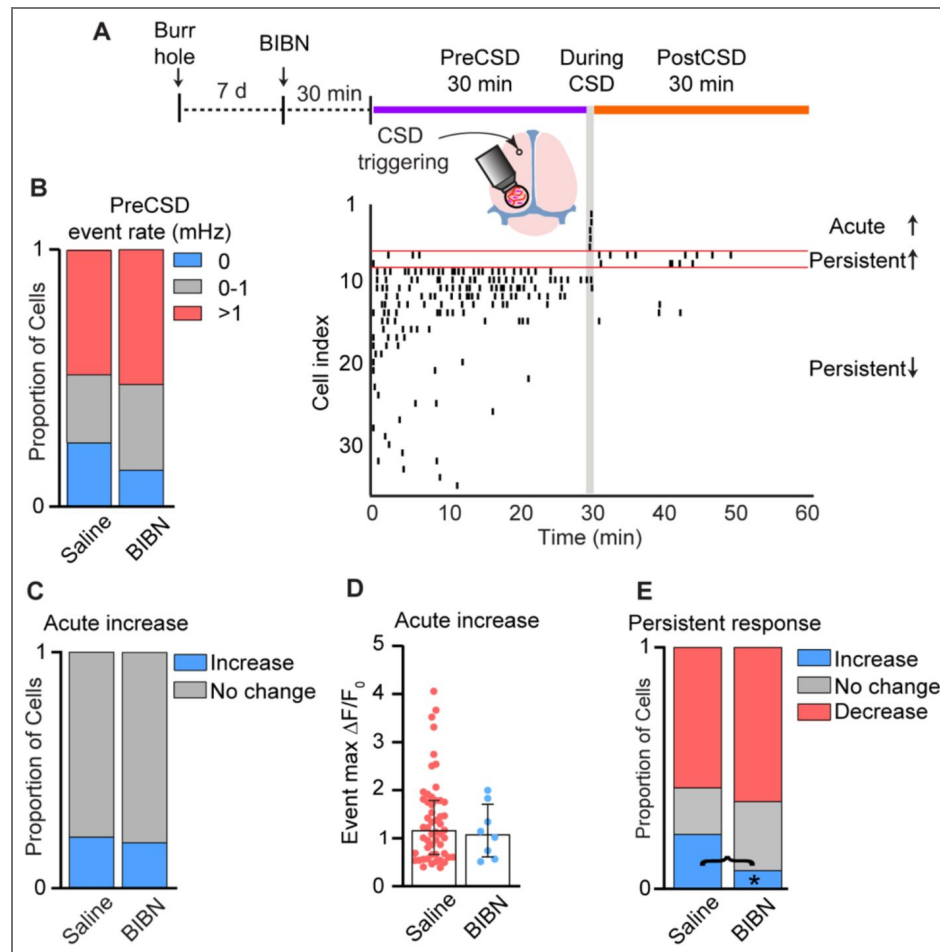


Figure 7. CGRP/RAMP1 signaling mediates CSD-related persistent increase in meningeal macrophage Ca^{2+} activity.

(A) Experimental setup and example data: (top) A small burr hole was drilled above the frontal cortex 7 days before Ca^{2+} imaging. Mice were pretreated with the RAMP1 antagonist BIBN4096 (BIBN) 30 min before imaging baseline macrophage Ca^{2+} activity (30 min, PreCSD). CSD was then induced with a pin prick, and macrophage Ca^{2+} activity was assessed during CSD (1 min during CSD), and post-CSD (30 min, post-CSD). (Bottom) A raster plot of macrophage Ca^{2+} activity showing the acute and persistent increases and persistent decrease in response to CSD. **(B)** RAMP1 inhibition does not affect the baseline (PreCSD) event rate. Data compared between macrophages imaged in saline-treated mice ($n=249$ cells) and BIBN-treated mice ($n=42$ cells). **(C)** RAMP1 antagonism does not affect the CSD-evoked acute increase in macrophage Ca^{2+} . Proportion of macrophages showing an acute response (increase vs. no change) in saline-treated mice ($n=249$ cells) and BIBN-treated mice ($n=42$ cells). **(D)** RAMP1 antagonism does not affect the magnitude of the acute macrophage Ca^{2+} signal. Event max $\Delta F/F_0$ in macrophages showing an acute Ca^{2+} increase in saline-treated mice ($n=53$ cells) and BIBN-treated mice ($n=8$ cells). **(E)** RAMP1 antagonism distinctly inhibits the persistent increase in macrophage Ca^{2+} activity post CSD. Proportion of macrophages showing persistent increase, persistent decrease, or no persistent change in saline-treated mice ($n=249$ cells) and BIBN-treated mice ($n=42$ cells). **(F)** Data (B, C, E) represents the proportion of cells. $*p<0.05$, Fisher's exact test. Data (D) are median $\pm$ IQR.

in dural perivascular macrophages tuned to behaviorally-driven dural vasomotion. Finally, we describe opposing Ca^{2+} responses of meningeal macrophages following CSD, and demonstrate the contribution of the CGRP/RAMP1 axis in mediating CSD-evoked persistent Ca^{2+} elevations.

The exact link between the distinct Ca^{2+} signal properties of meningeal macrophage subsets observed herein and their homeostatic function remains to be established. The lower event magnitude and noisier signal observed in dural perivascular macrophages may reflect functional interactions with the pulsation dynamics of dural vessels [46]. Indeed, by combining vascular and macrophage Ca^{2+} imaging, we demonstrate a tight temporal association between the diameter fluctuations of the dural vessels and the Ca^{2+} signal of their associated macrophages in awake, locomoting mice. Direct vascular-macrophage coupling may underlie this interaction, involving macrophages sensing vascular-related mechanical changes via Piezo1 signaling [60]. Functional vascular-macrophage interaction may also involve mural cells as inter-mediators [58, 61]. Studying whether macrophage Ca^{2+} signaling regulates dural vasomotion will require an experimental approach that has yet to be developed, allowing for their selective manipulation. The paucity of ongoing Ca^{2+} activity in perivascular macrophages situated in the leptomeninges we observed supports recent findings that subdural perivascular macrophages indirectly affect pial and parenchymal vasomotion via extracellular matrix remodeling [6].

Intracellular Ca^{2+} signal propagation has been observed in various non-excitabile cells, such as astrocytes [62]. We show that the majority of meningeal macrophages, including peri- and non-perivascular, exhibit intracellular Ca^{2+} signals that propagate throughout the entire cell, suggesting microdomain intracellular Ca^{2+} signal elevation, following release from internal stores. By characterizing the spatiotemporal relationships between Ca^{2+} signals in distinct cells, we also demonstrate synchronous events that are independent of spatial proximity, suggesting that synchronous Ca^{2+} activity is not driven by intercellular communication but rather by an external signal. Interestingly, our data indicate that synchronized events involve both peri- and non-perivascular macrophages, despite having distinct Ca^{2+} elevation features, suggesting that these meningeal macrophage subtypes similarly sense external signals.

Cortex-to-meninges signaling involves a relatively slow flow of soluble molecules in the cerebrospinal fluid reaching the subarachnoid space and advancing via arachnoid cuff exit points into the dura mater [11, 50]. Our finding of an acute Ca^{2+} elevation in a subset of extrasinusoidal perivascular dural macrophages coinciding with the CSD event suggests a rapid transfer of soluble signaling factors from the cortex across all meningeal layers [47]. Nevertheless, we cannot exclude a mechanically driven macrophage response to the acute meningeal deformation produced by the neuronal and glial swelling and shrinkage of the cortical extracellular space during CSD [53, 63, 64]. Our data also indicate a delayed and prolonged increase in Ca^{2+} signal in a relatively small subset of macrophages post-CSD, which could underlie their proinflammatory-like morphological change [49, 51]. Our findings also support the view that meningeal neuroimmune CGRP/RAMP1 axis serves as a mechanism responsible for this macrophage Ca^{2+} response, likely via the activation of their CGRP receptors [16]. Whether the relatively small subset of meningeal macrophages featuring increased Ca^{2+} signaling serves a protective role [65, 66] or a proinflammatory, destructive function [67] remains to be elucidated. Intriguingly, our data points to a persistent decrease in macrophage Ca^{2+} activity post-CSD, not involving CGRP/RAMP1 signaling, as the most prevalent response, raising the possibility that a hyperexcitable cortex dampens meningeal immunity.

Conclusions

We provide a detailed characterization of macrophage Ca^{2+} dynamics in homeostatic meninges, thereby expanding our understanding of their biological diversity. The coupling of dural perivascular macrophage Ca^{2+} signals and dural vasomotion may represent a unique homeostatic functional dural macrophage-vascular unit that controls dural perfusion. The diversity of meningeal macrophage Ca^{2+} responses to CSD further highlights the complexity of brain-to-meninges neuroimmune signaling and meningeal macrophage function in neurological disorders

such as migraine, traumatic brain injury, and stroke. Our study also provides essential genetic and data analysis tools to further understand the molecular signaling underlying macrophage function at steady state and neuropathological conditions.

Methods

Animals

All experimental procedures were approved by the Beth Israel Deaconess Medical Center Institutional Animal Care and Use Committee. Experiments were conducted on adult $\text{Pf4}^{\text{Cre/WT}};\text{TIGRE2.0}^{\text{GCaMP6s/WT}} \text{Ca}^{2+}$ reporter mice (8–17 weeks) (9 males, 5 females). Mice were generated by crossing Pf4^{Cre} (C57BL/6-Tg (Pf4-icre) Q3Rsko/J, Jackson laboratory, Strain #008535) mice with $\text{GCaMP6s}^{\text{fl/fl}}$ (Ai162D; B6.Cg-Igs7^{tm162.1(tetO-GCaMP6s,CAG-tTA2)Hze/J}, Jackson laboratory, Strain #031562) mice. Animals were genotyped by Transnetyx Inc.

Surgical procedures

Animals were anesthetized using isoflurane in 100% O_2 (induction: 3%; maintenance: 1.5–2%) and placed on a heating pad with a rectal probe attached to a stereotaxic frame to monitor animal body temperature during surgery. Animals received dexamethasone (8 mg/kg, i.p.) and Meloxicam SR (4 mg/kg, s.c.) to reduce inflammation and improve surgical outcomes. An eye ointment was used to prevent ocular drying. Mice were implanted with a titanium headpost and a 3 mm glass cranial window (1.5 mm lateral and 2 mm posterior to Bregma) over an intact dura covering the left posterior neocortex [53]. Immediately after surgery, the mouse cage was placed on a water-circulating heating pad for faster recovery. Animals were then single-housed with access to a running wheel and a hut and allowed to recover for at least one week.

Wheel running acclimation

After the cranial window surgery, mice were allowed to recover for at least a week. To reduce stress associated with head-fixation during imaging and habituate to wheel running, the mice received multiple training sessions (10 min to 1 h over 3–4 days). In each session, the mouse was placed on a 3D printed running wheel, with its headpost attached to two clamps, and allowed to locomote freely.

Two-photon Ca^{2+} imaging

Awake-behaving mice were head-fixed to the running wheel by its headpost (Figure 1). We used a two-photon microscope (Neurolabware) with a Nikon 16X, 0.8 NA objective to acquire images at 15.5 Hz with digital zoom set at 4X (312 x 212 μm^2 FOV). A MaiTai laser set to 920 nm with 25–40 mW power was used to excite fluorescence. The Scanbox package for MATLAB (Neurolabware) was used to control the microscope and acquire images and wheel running data. To image the meningeal vasculature, mice were administered 70kDa TRITC-Dextran tracer (50 mg/kg, i.v.; Sigma-Aldrich).

Behavioral tracking during two-photon imaging

We recorded running speed in MATLAB using a custom-made encoder (Arduino) coupled to the 3D-printed running wheel. See below for details of analyses of behavioral variables.

Induction of CSD and pharmacological treatment

For CSD induction, a 1 mm burr hole was drilled at the frontal bone (1.5 mm anterior to the cranial window) to allow access to the brain cortical surface. A small amount of silicone elastomer (Kwik-Cast, WPI) was placed to cover the burr hole opening, and the animal was left to recover for at least one week. CSD was induced using a brief 2-second cortical pinprick [53]. CSD induction was confirmed by the identification of a short-lasting meningeal deformation and/or transient pial

constriction [53]. On the experimental day, 30 minutes before baseline recording, mice were pretreated with the CGRP/RAMP1 antagonist BIBN4096 (0.3 ml, 1 mg/kg, i.p., Tocris) [54] or 0.3 ml of saline (Vehicle control).

Quantification and statistical analysis

Two-photon imaging movie processing

We used a discrete Fourier transform to perform rigid registration to correct translation changes due to brain motion during locomotion. Movies were then downsampled to 1.03 Hz. Locomotion signals were detected as described [41]. All image processing and locomotion signal extraction were performed in MATLAB 2021b (Mathworks).

Ca²⁺ signal detection pipeline

For detecting macrophage Ca²⁺ signals, we used the Activity Quantification and Analysis (AQuA2) platform that implements an event-based approach with advanced machine learning techniques for temporal and spatial segmentation of Ca²⁺ fluorescence events³⁸. Importantly, this computational platform captures event dynamics beyond traditional ROI-based approaches. The following user-defined parameters were input: 0.49 μm/pixel spatial resolution, 1.03 Hz temporal resolution, 1-second minimal event duration detection; window of event size between 25% (125 μm²/ 523 pixels) and 150% (750 μm²/ 3140 pixels). Every event and cell identified was followed by a manual visual check. Subsequently, each cell was labelled as perivascular or non-perivascular according to its location relative to vessels. Morphological features (area, perimeter, circularity) and spatiotemporal aspects of the Ca²⁺ signals (i.e., ΔF/F₀ dynamics, frequency, amplitude, duration) were used to analyze cell-specific characteristics and Ca²⁺ activity profiles. We employed the AQuA2 automatically-generated function [42] to characterize intracellular Ca²⁺ propagation. Intercellular Ca²⁺ activity was evaluated by analyzing temporally co-occurring events (synchronized event pairs), as well as their corresponding spatial localization in the FOV. All post-processing of AQuA2-generated data was performed using MATLAB 2021b (Mathworks).

Clustering of Ca²⁺ dynamics

We used a Savitzky-Golay filter to detrend and smooth Ca²⁺ activity traces. Polynomial order and frame size were optimized for each cell by selecting the combination of parameters that yielded a better signal-to-noise ratio. From the filtered signals, we extracted the dominant frequency using Fast Fourier Transform, and peak counts using minimum peak prominence (threshold of 10% signal amplitude range). The optimal number of clusters was determined using the Elbow method, after which k-means clustering was applied to group cells based on their signal characteristics.

Analysis of CSD-related changes in macrophage Ca²⁺ dynamics

For analyzing the effects of CSD on macrophages' Ca²⁺ dynamics, we divided the activity of each cell into three phases: 'PreCSD' (minutes 0 – 30), 'DuringCSD' (minutes 30 – 31), and 'PostCSD' (minutes 31 – 61). Ca²⁺ event rate during CSD and PostCSD was compared against that of the PreCSD baseline for evaluation of acute and persistent Ca²⁺ responses, respectively. Ca²⁺ responses were categorized as increased (event rate > 2x PreCSD), decreased (event rate < 0.5x PreCSD), or unchanged.

Analysis of locomotion

We extracted running speed (cm/sec) from the wheel encoder. To infer the locomotion state, we first concatenated all velocity signals obtained from a given mouse across all experiments and trained a two-state Hidden Markov Model using the MATLAB function 'hmmtrain'. Then, the locomotion state was inferred for each individual imaging run by applying the MATLAB function 'hmmviterbi' with the model trained on the concatenated data. Locomotion bouts were defined as periods when the locomotion state was sustained for at least two seconds.

Vascular signals

We calculated changes in vascular diameter by first generating a maximum intensity projection of the red channel and drawing polygons (ROIs) around each vessel. Subsequently, for each frame, pixels inside each ROI were extracted, and a Radon transform was applied to get a 1D vessel profile. Radon transform of the first frame was used as a reference to normalize the results of all subsequent frames, resulting in a time series of normalized diameter traces.

General linear models (GLM)

We investigate the functional interaction between dural perivascular macrophage Ca^{2+} activity and dural vasomotion by fitting a Gaussian GLMs using the GLMnet package (MATLAB 2021b), with elastic net regularization ($\alpha=0.01$) and ten-fold cross-validation. We employed a two-step modeling. The first model was used to identify dural vessels by evaluating the correlation of vessel diameter changes to locomotion state, classifying dural or pial vessels according to their vasoconstriction and vasodilation dynamics, respectively [46]. The second modeling step evaluated the relative contribution of perivascular dural Ca^{2+} activity signals to dural vessel diameter changes. All signals were downsampled to 1.03 Hz to match the frame rate used when extracting fluorescence Ca^{2+} signals in AQuA2. To allow potential anticipatory or delayed responses of diameter- to-locomotion or Ca^{2+} fluorescence-to-diameter, the time interval used to analyze the response to the predictor was set from -60 s to +60 s. The GLM was trained on 75% of the data, and all predictions and model performances reported are from the remaining 25% testing set. A threshold of 0.1 goodness-of-fit deviance explained was set. For each predictor temporal shift, a response coefficient was generated. Centroid delay between predictor and response was calculated, weighted by the absolute value of the coefficient (i.e., center-of-mass, COM).

Statistical analysis

All statistical analysis was performed using GraphPad Prism 10.4 and MATLAB 2021b. Data were analyzed using a Wilcoxon matched-pairs signed rank sum test or a Mann-Whitney U-test. Distribution of categorical data was analyzed using the Chi-square or Fisher's exact test. P-values are indicated as follows: * $p<0.05$, ** $p<0.01$, *** $p<0.001$, **** $p<0.0001$.

Additional information

Data availability

All data needed to evaluate the conclusions in the paper are present in the paper. The code used for analyzing the data in this study was deposited in the Levy Lab GitHub account.

Code for movie processing is available at:

<https://github.com/levylabheadache/MovieProcessing/tree/SCN> 

Code for locomotion analysis is available at:

<https://github.com/levylabheadache/Locomotion/tree/SCN> 

Code for post Aqua2-processing is available at:

<https://github.com/levylabheadache/Aqua2Processing> 

Code for vascular segmentation is available at:

<https://github.com/levylabheadache/Vasculature/tree/SCN> 

Code for GLM is available at:

https://github.com/levylabheadache/GeneralLinearModel_Macrophages 

Any additional information is available from the corresponding author upon request.

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Authors information

D.L. conceived the project. S.C-N and D.L. wrote the manuscript. S.C-N performed two-photon imaging and data analysis with help from D.L. S.C-N, C.W., and A.G. performed surgeries.

Additional files

Video S1. [↗](#) Macrophage Ca^{2+} dynamics in homeostatic meninges of awake mice. Example two-photon imaging showing Ca^{2+} activity in perivascular and non-perivascular meningeal macrophages.

Video S2. [↗](#) Acute and persistent changes in meningeal macrophage Ca^{2+} activity in response to CSD. Example two-photon imaging of meningeal macrophage Ca^{2+} activity at baseline, during, and following CSD. The arrow indicates a macrophage showing an acute Ca^{2+} elevation, the arrowhead depicts a delayed and persistent Ca^{2+} elevation, and the asterisk, a macrophage showing a persistent decreased Ca^{2+} activity.

Abbreviations

AQuA2: Activity Quantification and Analysis
 Ca^{2+} : Cytoplasmic calcium
 CGRP: Calcitonin gene-related peptide
 CSD: Cortical spreading depolarization (depression)
 CX3CR1: CX3C Motif chemokine receptor 1
 GLM: General Linear Model
 NP: Non-perivascular
 P: Perivascular
 Pf4: Platelet factor 4
 RAMP1: Receptor activity modifying protein 1
 TRITC: Tetramethylrhodamine isothiocyanate

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Peer reviews

Reviewer #1 (Public review):

Summary:

This study presents a technically sophisticated intravital two-photon calcium imaging approach to characterize meningeal macrophage Ca^{2+} dynamics in awake mice. The development of a Pf4Cre:GCaMP6s reporter line and the integration of event-based Ca^{2+} analysis represent clear methodological strengths. The findings reveal niche-specific Ca^{2+} signaling patterns and heterogeneous macrophage responses to cortical spreading depolarization (CSD), with potential relevance to migraine and neuroinflammatory conditions. Despite these strengths, several conceptual, technical, and interpretational issues limit the impact and mechanistic depth of the study. Addressing the points below would substantially strengthen the manuscript.

Strengths:

The use of chronic two-photon Ca^{2+} imaging in awake, behaving mice represents a major technical strength, minimizing confounds introduced by anesthesia. The development of a Pf4Cre:GCaMP6s reporter line, combined with high-resolution intravital imaging, enables long-term and subcellular analysis of macrophage Ca^{2+} dynamics in the meninges.

The comparison between perivascular and non-perivascular macrophages reveals clear niche-dependent differences in Ca^{2+} signaling properties. The identification of macrophage Ca^{2+} activity temporally coupled to dural vasomotion is particularly intriguing and highlights a potential macrophage-vascular functional unit in the dura.

By linking macrophage Ca^{2+} responses to CSD and implicating CGRP/RAMP1 signaling in a subset of these responses, the study connects meningeal macrophage activity to clinically relevant neuroimmune pathways involved in migraine and other neurological disorders.

Weaknesses:

The manuscript relies heavily on Pf4Cre-driven GCaMP6s expression to selectively image meningeal macrophages. Although prior studies are cited to support Pf4 specificity, Pf4 is not an exclusively macrophage-restricted marker, and developmental recombination cannot be excluded. The authors should provide direct validation of reporter specificity in the adult meninges (e.g., co-labeling with established macrophage markers and exclusion of other Pf4-expressing lineages). At minimum, the limitations of Pf4Cre-based labeling should be discussed more explicitly, particularly regarding how off-target expression might affect Ca^{2+} signal interpretation.

The manuscript offers an extensive characterization of Ca^{2+} event features (frequency spectra, propagation patterns, synchrony), but the biological significance of these signals is largely speculative. There is no direct link established between Ca^{2+} activity patterns and macrophage function (e.g., activation state, motility, cytokine release, or interaction with other meningeal components). The discussion frequently implies functional specialization based on Ca^{2+} dynamics without experimental validation. To strengthen the conceptual impact, a clearer framing of the study as a foundational descriptive resource, rather than a functional dissection, would improve alignment between data and conclusions.

The GLM analysis revealing coupling between dural perivascular macrophage Ca^{2+} activity and vasomotion is technically sophisticated and intriguing. However, the directionality of this relationship remains unresolved. The current data do not distinguish whether macrophages actively regulate vasomotion, respond to mechanical or hemodynamic changes, or are co-

modulated by neural activity. Statements suggesting that macrophages may "mediate" vasomotion are therefore premature. The authors should reframe these conclusions more cautiously, emphasizing correlation rather than causation, and expand the discussion to explicitly outline experimental strategies required to establish causality (e.g., macrophage-specific Ca^{2+} manipulation).

The authors conclude that synchronous Ca^{2+} events across macrophages are driven by extrinsic signals rather than intercellular communication, based primarily on distance-time analyses. This conclusion is not sufficiently supported, as spatial independence alone does not exclude paracrine signaling, vascular cues, or network-level coordination. No perturbation experiments are presented to test alternative mechanisms. The authors can either provide additional experimental evidence or rephrase the conclusion to acknowledge that the source of synchrony remains unresolved.

A major and potentially important finding is that the dominant macrophage response to CSD is a persistent decrease in Ca^{2+} activity, which is independent of CGRP/RAMP1 signaling. However, this phenomenon is not mechanistically explored. It remains unclear whether Ca^{2+} suppression reflects macrophage inhibition, altered viability, homeostatic resetting, or an anti-inflammatory program. Minimally, the discussion should be more deeply engaged with possible interpretations and implications of this finding.

The pharmacological blockade of RAMP1 supports a role for CGRP signaling in persistent Ca^{2+} increases after CSD, but the experiments are based on a relatively small number of cells and animals. The limited sample size constrains confidence in the generality of the conclusions. Pharmacological inhibition alone does not establish cell-autonomous effects in macrophages. The authors should acknowledge these limitations more explicitly and avoid overextension of the conclusions.

<https://doi.org/10.7554/eLife.109888.1.sa3>

Reviewer #2 (Public review):

Using chronic intravital two-photon imaging of calcium dynamics in meningeal macrophages in Pf4Cre:TIGRE2.0-GCaMP6 mice, the study identified heterogeneous features of perivascular and non-perivascular meningeal macrophages at steady state and in response to cortical spreading depolarization (CSD). Analyses of calcium dynamics and blood vessels revealed a subpopulation of perivascular meningeal macrophages whose activity is coupled to behaviorally driven diameter fluctuations of their associated vessels. The analyses also investigated synchrony between different macrophage populations and revealed a role for CGRP/RAMP1 signaling in the CSD-induced increase, but not the decrease, in calcium transients.

This is a timely study at both the technical and conceptual levels, examining calcium dynamics of meningeal macrophages in vivo. The conclusions are well supported by the findings and will provide an important foundation for future research on immune cell dynamics within the meninges in vivo. The paper is well written and clearly presented.

I have only minor comments.

(1) Please indicate the formal definition of perivascular versus non-perivascular macrophages in terms of distance from the blood vessel. This information is not provided in the main text or the Methods. In addition, please explain how the meningeal vasculature was imaged in the main text.

(2) Similarly, the method used to induce acute CSD (pin prick) is not described in the main text and is only mentioned in the figure legends and Methods. Additional background on the

neurobiology of acute CSD, as well as the resulting brain activity and neuroinflammatory responses, could be helpful.

<https://doi.org/10.7554/eLife.109888.1.sa2>

Reviewer #3 (Public review):

Summary:

The authors of this report wish to show that distinct populations of meningeal macrophages respond to cortical spreading depolarization (CSD) via unique calcium activity patterns depending on their location in the meningeal sub-compartments. Perivascular macrophages display calcium signaling properties that are sometimes in opposition to non-perivascular macrophages. Many of the meningeal macrophages also displayed synchronous activity at variable distances from one another. Other macrophages were found to display calcium signals in response to dural vasomotion. CSD could induce variable calcium responses in both perivascular and non-perivascular macrophages in the meninges, in part due to RAMP1-dependent effects. Results will inform future research on the calcium responses displayed by macrophages in the meninges under both normal and pathological conditions.

Strengths:

Sophisticated in vivo imaging of meningeal immune cells is employed in the study, which has not been performed previously. A detailed analysis of the distinct calcium dynamics in various subtypes of meningeal macrophages is provided. Functional relevance of the responses is also noted in relation to CSD events.

Weaknesses:

The specificity of the methods used to target both meningeal macrophages and RAMP1 is limited. Additional discussion points on the functional relevance of the two subtypes of meningeal macrophages and their calcium responses are warranted. A section on potential pitfalls should be included.

<https://doi.org/10.7554/eLife.109888.1.sa1>

Author response:

Public Reviews:

Reviewer #1 (Public review):

Strengths:

(1) *The use of chronic two-photon Ca^{2+} imaging in awake, behaving mice represents a major technical strength, minimizing confounds introduced by anesthesia. The development of a Pf4Cre:GCaMP6s reporter line, combined with high-resolution intravital imaging, enables long-term and subcellular analysis of macrophage Ca^{2+} dynamics in the meninges.*

(2) *The comparison between perivascular and non-perivascular macrophages reveals clear niche-dependent differences in Ca^{2+} signaling properties. The identification of macrophage Ca^{2+} activity temporally coupled to dural vasomotion is particularly intriguing and highlights a potential macrophage-vascular functional unit in the dura.*

(3) *By linking macrophage Ca^{2+} responses to CSD and implicating CGRP/RAMP1 signaling in a subset of these responses, the study connects meningeal macrophage activity to*

clinically relevant neuroimmune pathways involved in migraine and other neurological disorders.

Thank you for recognizing the strengths in our work.

Weaknesses:

(1) The manuscript relies heavily on Pf4Cre-driven GCaMP6s expression to selectively image meningeal macrophages. Although prior studies are cited to support Pf4 specificity, Pf4 is not an exclusively macrophage-restricted marker, and developmental recombination cannot be excluded. The authors should provide direct validation of reporter specificity in the adult meninges (e.g., co-labeling with established macrophage markers and exclusion of other Pf4-expressing lineages). At minimum, the limitations of Pf4Cre-based labeling should be discussed more explicitly, particularly regarding how off-target expression might affect Ca²⁺ signal interpretation.

We acknowledge that PF4 is not an exclusively macrophage-restricted marker. Yet, among meningeal immunocytes, it is almost exclusively expressed in macrophages (1, 2). Furthermore, in the adult mouse meninges, Pf4^{Cre}-based reporter lines label nearly all dural and leptomeningeal macrophages and almost no other cells (3, 4). This Cre line has also been used to target border-associated macrophages (2, 4). Moreover, a recent study suggests that the bacterial artificial chromosome used to generate the Pf4^{Cre} line does not affect meningeal macrophage activity (4). Nonetheless, while we already discussed PF4 expression in meningeal megakaryocytes, in a revised version, we plan to discuss the possibility that a very small population of other meningeal immune cells may also be labeled.

(2) The manuscript offers an extensive characterization of Ca²⁺ event features (frequency spectra, propagation patterns, synchrony), but the biological significance of these signals is largely speculative. There is no direct link established between Ca²⁺ activity patterns and macrophage function (e.g., activation state, motility, cytokine release, or interaction with other meningeal components). The discussion frequently implies functional specialization based on Ca²⁺ dynamics without experimental validation. To strengthen the conceptual impact, a clearer framing of the study as a foundational descriptive resource, rather than a functional dissection, would improve alignment between data and conclusions.

In our discussion, we indicated that “the exact link between the distinct Ca²⁺ signal properties of meningeal macrophage subsets observed herein and their homeostatic function remains to be established”. In a revised version, we plan to further acknowledge that this is primarily a descriptive study that provides a foundational landscape of Ca²⁺ dynamics in meningeal macrophages.

(3) The GLM analysis revealing coupling between dural perivascular macrophage Ca²⁺ activity and vasomotion is technically sophisticated and intriguing. However, the directionality of this relationship remains unresolved. The current data do not distinguish whether macrophages actively regulate vasomotion, respond to mechanical or hemodynamic changes, or are co-modulated by neural activity. Statements suggesting that macrophages may “mediate” vasomotion are therefore premature. The authors should reframe these conclusions more cautiously, emphasizing correlation rather than causation, and expand the discussion to explicitly outline experimental strategies required to establish causality (e.g., macrophage-specific Ca²⁺ manipulation).

In the results section, we indicated that our data suggest that dural perivascular macrophages are functionally coupled to locomotion-driven dural vasomotion, either responding to it or mediating it. Furthermore, in our discussion, we discussed the possibilities that 1) macrophages sense vascular-related mechanical changes and 2) macrophage Ca²⁺ signaling may regulate dural vasomotion. Moreover, we explicitly state that studying causality will

require an experimental approach that has yet to be developed, enabling selective manipulation of dural perivascular macrophages.

(4) The authors conclude that synchronous Ca^{2+} events across macrophages are driven by extrinsic signals rather than intercellular communication, based primarily on distance-time analyses. This conclusion is not sufficiently supported, as spatial independence alone does not exclude paracrine signaling, vascular cues, or network-level coordination. No perturbation experiments are presented to test alternative mechanisms. The authors can either provide additional experimental evidence or rephrase the conclusion to acknowledge that the source of synchrony remains unresolved.

Thank you for this suggestion. In the revision, we will indicate that the source of synchrony remains unresolved.

(5) A major and potentially important finding is that the dominant macrophage response to CSD is a persistent decrease in Ca^{2+} activity, which is independent of CGRP/RAMP1 signaling. However, this phenomenon is not mechanistically explored. It remains unclear whether Ca^{2+} suppression reflects macrophage inhibition, altered viability, homeostatic resetting, or an anti-inflammatory program. Minimally, the discussion should be more deeply engaged with possible interpretations and implications of this finding.

While we propose that the decrease in macrophage calcium signaling following CSD could indicate that a hyperexcitable cortex dampens meningeal immunity, in the revised version, we plan to elaborate on the possible implications of this finding.

(6) The pharmacological blockade of RAMP1 supports a role for CGRP signaling in persistent Ca^{2+} increases after CSD, but the experiments are based on a relatively small number of cells and animals. The limited sample size constrains confidence in the generality of the conclusions. Pharmacological inhibition alone does not establish cell-autonomous effects in macrophages. The authors should acknowledge these limitations more explicitly and avoid overextension of the conclusions.

We plan to acknowledge these limitations.

Reviewer #2 (Public review):

Using chronic intravital two-photon imaging of calcium dynamics in meningeal macrophages in Pf4Cre:TIGRE2.0-GCaMP6 mice, the study identified heterogeneous features of perivascular and non-perivascular meningeal macrophages at steady state and in response to cortical spreading depolarization (CSD). Analyses of calcium dynamics and blood vessels revealed a subpopulation of perivascular meningeal macrophages whose activity is coupled to behaviorally driven diameter fluctuations of their associated vessels. The analyses also investigated synchrony between different macrophage populations and revealed a role for CGRP/RAMP1 signaling in the CSD-induced increase, but not the decrease, in calcium transients.

This is a timely study at both the technical and conceptual levels, examining calcium dynamics of meningeal macrophages in vivo. The conclusions are well supported by the findings and will provide an important foundation for future research on immune cell dynamics within the meninges in vivo. The paper is well written and clearly presented.

Thank you.

I have only minor comments.

(1) Please indicate the formal definition of perivascular versus non-perivascular macrophages in terms of distance from the blood vessel. This information is not provided

in the main text or the Methods. In addition, please explain how the meningeal vasculature was imaged in the main text.

We did not measure the exact distance of the perivascular macrophages from the blood vessels, but defined them as such based on previous data showing that these cells reside along the abluminal surface and maintain tight interactions with mural cells (5). We plan to provide this information in the revised manuscript.

(2) Similarly, the method used to induce acute CSD (pin prick) is not described in the main text and is only mentioned in the figure legends and Methods. Additional background on the neurobiology of acute CSD, as well as the resulting brain activity and neuroinflammatory responses, could be helpful.

We plan to add the method for inducing CSD (i.e., a pinprick in the frontal cortex) to the Results section and provide more background in the Introduction section.

Reviewer #3 (Public review):

Strengths:

Sophisticated in vivo imaging of meningeal immune cells is employed in the study, which has not been performed previously. A detailed analysis of the distinct calcium dynamics in various subtypes of meningeal macrophages is provided. Functional relevance of the responses is also noted in relation to CSD events.

Thank you for recognizing the strengths of our paper

Weaknesses:

(1) The specificity of the methods used to target both meningeal macrophages and RAMP1 is limited. Additional discussion points on the functional relevance of the two subtypes of meningeal macrophages and their calcium responses are warranted. A section on potential pitfalls should be included.

We plan to address these issues in the revision

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